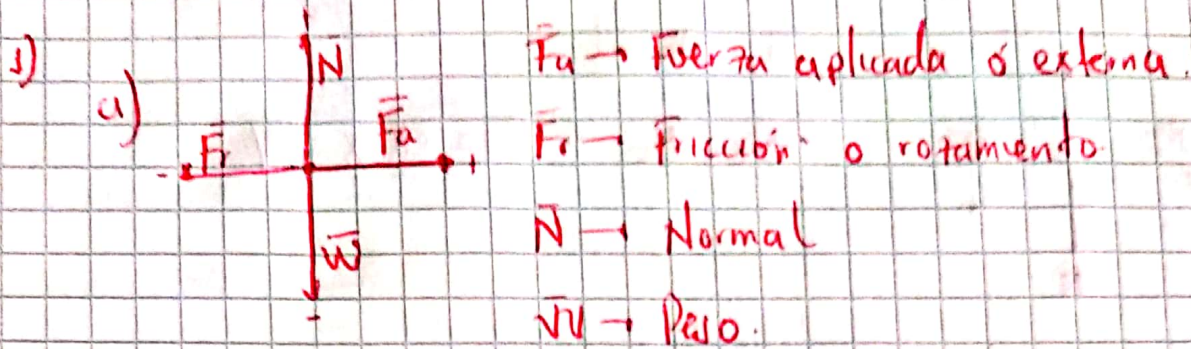


Clave JJ F05 e (D).



b) $\sum F_y = N - W = 0$

$$N = W$$

$$N = mg$$

$$N = 12 \text{ Kg} \cdot 9.8 \text{ m/s}^2$$

$$\boxed{N = 117.6 \text{ N.}}$$

c) $W = mg$

$$\boxed{W = 117.6 \text{ N}}$$

d) $\sum F_x = F_a - F_r = 0$

$$F_a = F_r$$

$$F_a = \mu \cdot N$$

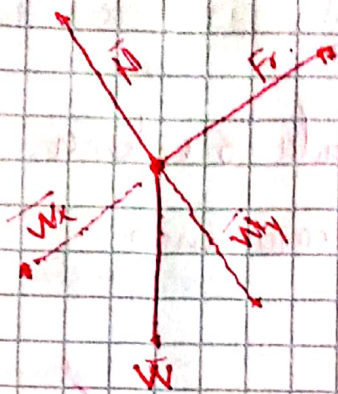
$$\mu = \frac{F_a}{N}$$

$$\mu = \frac{450 \text{ N}}{117.6 \text{ N}}$$

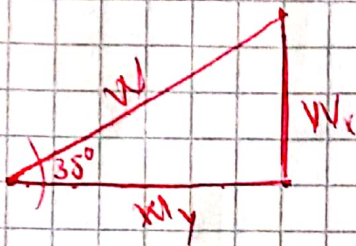
$$\boxed{\mu = 3.8}$$

2)

a)

 $\vec{F}_r \rightarrow$ Fuerza de rozamiento $\vec{N} \rightarrow$ Normal $W_x, W_y \rightarrow$ Peso en x y y.

b)



$$W_x = W \cdot \sin \theta$$

$$W_y = W \cdot \cos \theta$$

$$W = m \cdot g$$

$$W = 8 \times 9.8 \text{ m/s}^2$$

$$\boxed{W = 78.4 \text{ N}}$$

$$W_x = mg \sin \theta$$

$$W_x = 8 \times 9.8 \cdot \sin 35$$

$$\boxed{W_x = 44.96 \text{ N}}$$

$$W_y = mg \cos \theta$$

$$W_y = 8 \times 9.8 \cos 35$$

$$\boxed{W_y = 64.2 \text{ N}}$$

$$c) \sum F_y = 0$$

$$\vec{N} - \vec{W}_y = 0$$

$$\vec{N} = \vec{W}_y$$

$$\boxed{\vec{N} = 64.2 \text{ N}}$$

$$d) \vec{F}_r = \mu \cdot \vec{N}$$

$$\vec{F}_r = 0.15 \cdot 64.2 \text{ N}$$

$$\boxed{\vec{F}_r = 9.63 \text{ N}}$$

$$e) \sum F_x = m \cdot a$$

$$F_r - W_x = -m \cdot a$$

$$\frac{F_r - W_x}{-m} = a$$

$$a = \frac{9.63 \text{ N} - 44.96 \text{ N}}{-8 \text{ Kg}}$$

$$a = \frac{-35.33 \text{ N}}{-8 \text{ Kg}}$$

$$a = 4.41 \frac{\text{N} (\text{Kg} \cdot \text{m/s}^2)}{\text{Kg} (\text{Kg})}$$

$$\boxed{a = 4.41 \text{ m/s}^2}$$

- 3) El estudiante deberá asociar la primera y segunda ley de Newton a la situación como mínimo.
- es decir:
- Considerar el objeto en reposo
 - Considerar el objeto en movimiento

- 4) a) F
- b) V